

5G RAN

SA Mobility Management in Idle Mode Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Deleted the "SMTC Measurement Configuration" section, and moved its content to [SMTC Measurement Configuration](#) under [4.1.3.3 Neighboring Cell Measurement for Cell Reselection](#).
Added descriptions of when the T320 timer starts. For details, see [4.1.3.1.2 Dedicated Cell Reselection Priority](#).

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-04-27).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for the configurable T320 timer. For details, see: • 4.1.3.1.2 Dedicated Cell Reselection Priority • 4.2.2 Impacts • 4.3.2.3 Function Impacts • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Added parameters: • NRCellReselConfig.T320ForOther • NRCellReselConfig.T320ForMlb	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Optimized the mechanism for configuring the SMTC offset. For details, see SMTC Measurement Configuration under 4.1.3.3 Neighboring Cell Measurement for Cell Reselection.	Added the NRCellFreqRelation.SaSmTcOffset parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added switch control for UE-layering-based differentiated cell reselection priority configuration. For details, see: • 4.1.3.1.2 Dedicated Cell Reselection Priority • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Added the NRCellMobilityConfig.IdleReselPolicySw parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

Moved descriptions of the impact relationships—specifically, the relationships between dedicated cell reselection priorities and the following functions: "inter-frequency handover policy for voice service-capable UEs" and "voice-

capability-based inter-frequency handover back to the HPLMN"—from [4.1.3 Cell Reselection](#) to [4.3.2.3 Function Impacts](#).

Reorganized descriptions of cell reselection priorities by distinguishing between common and dedicated cell reselection priorities, and revised descriptions of dedicated cell reselection priorities. For details, see [4.1.3 Cell Reselection](#), [4.1.3.1.1 Common Cell Reselection Priority](#), and [4.1.3.1.2 Dedicated Cell Reselection Priority](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the names of reference documents.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010014	Mobility Management	4 Mobility Management in Idle Mode

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Mobility management in idle mode	In non-standalone (NSA) networking, UEs in idle mode camp on LTE cells. As such, standalone (SA) mobility management in idle mode differs from NSA mobility management in idle mode. This document describes SA mobility management in idle mode. For details about NSA mobility management in idle mode, see <i>Idle Mode Management</i> in <i>eRAN Feature Documentation</i> .	4 Mobility Management in Idle Mode

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Mobility management in idle mode	Supported only in low frequency bands	4 Mobility Management in Idle Mode

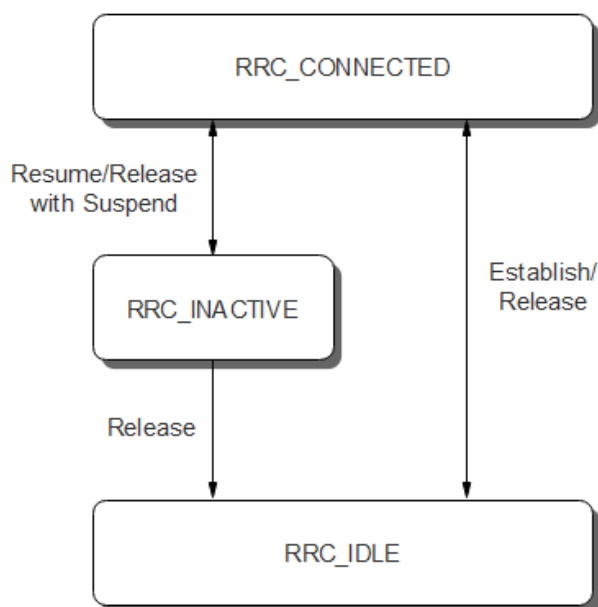
3 Overview

Mobility management is a basic function used to ensure that services stay uninterrupted as UEs move through mobile networks.

3.1 RRC State

In a 5G network, a UE at any time is in one of the following radio resource control (RRC) states: RRC_CONNECTED, RRC_IDLE, and RRC_INACTIVE states, as presented in [Figure 3-1](#).

Figure 3-1 RRC states



- RRC_CONNECTED: An RRC connection is established between the UE and the gNodeB.
- RRC_IDLE: No RRC connection is established between the UE and the gNodeB.
- RRC_INACTIVE: In this state, the UE suspends data processing. However, the gNodeB still maintains the UE's context information, and as a result, the core network considers that the UE remains in the CM-CONNECTED state (as if it

were still in the RRC_CONNECTED state). When data transmission is required, the UE can quickly transit to the RRC_CONNECTED state.

 NOTE

RRC connection release is performed for transition from RRC_CONNECTED to RRC_IDLE or from RRC_CONNECTED to RRC_INACTIVE. The state (RRC_IDLE or RRC_INACTIVE) that a UE has entered can be checked based on whether *suspendConfig* is included in the RRCRelease message. If *suspendConfig* is included in the RRCRelease message, the UE has entered the RRC_INACTIVE state. If *suspendConfig* is not included in the RRCRelease message, the UE has entered the RRC_IDLE state.

For details about the RRC states, see section 4.2.1 "UE states and state transitions including inter RAT" in 3GPP TS 38.331 V15.5.1.

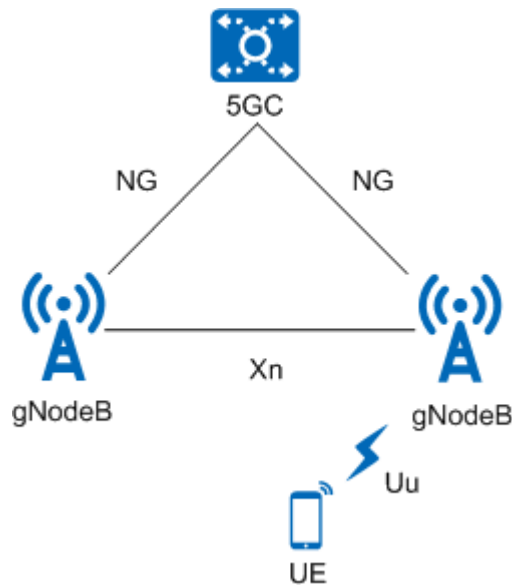
3.2 Mobility Management Scenarios

5G supports SA networking and NSA networking. Based on networking scenarios, mobility management is categorized into SA mobility management, NSA mobility management, and NSA and SA hybrid mobility management. **This document zooms in on functions related to SA mobility management in idle mode.**

SA Mobility Management

SA networking uses an end-to-end 5G network architecture, where the terminals, base stations, and core network comply with 5G specifications. [Figure 3-2](#) shows SA networking. For details about SA networking, see *5G Networking and Signaling*.

Figure 3-2 SA networking (Option 2 as an example)



In SA networking, mobility management is classified into intra-RAT mobility management and inter-RAT mobility management, depending on whether a serving cell changes between two networks of different radio access technologies (RATs). For details about SA inter-RAT mobility management, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*, *Interoperability Between E-*

UTRAN and NG-RAN in Idle Mode, and NSA-SA Selection Based on User Experience.

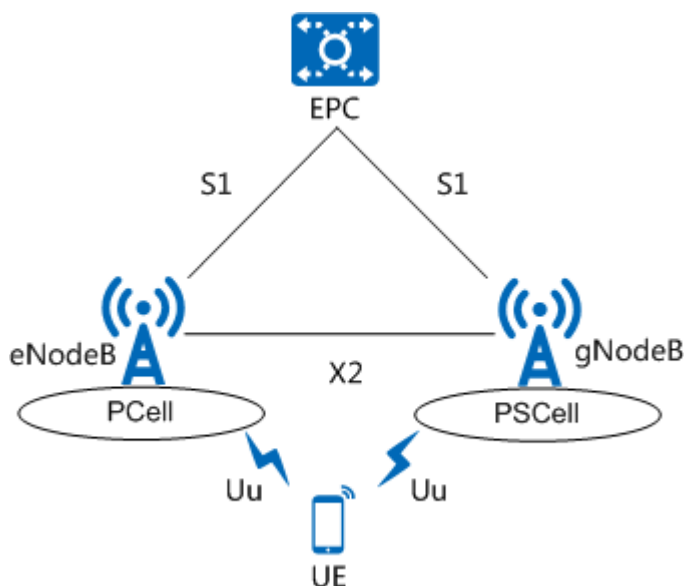
Based on the RRC states of UEs, intra-RAT mobility management in SA networking is categorized as follows:

- Mobility management in connected mode (RRC_CONNECTED state). For details, see *SA Mobility Management in Connected Mode*.
- Mobility management in idle mode (RRC_IDLE state), which is described in this document.
- Mobility management in inactive mode (RRC_INACTIVE state). For details, see *SA Mobility Management in Inactive Mode*.

NSA Mobility Management

In NSA networking, terminals supporting E-UTRA-NR dual connectivity (EN-DC) connect to both an LTE base station and a 5G New Radio (NR) base station, using the radio resources of both of the base stations for transmission. The cell served by the LTE base station (eNodeB) is the primary cell (PCell), and the cell served by the 5G NR base station (gNodeB) is the primary SCG cell (PSCell). **Figure 3-3** shows the NSA EN-DC networking. For details about NSA networking, see *EPC-based NSA Basics*.

Figure 3-3 NSA EN-DC networking (Option 3x as an example)



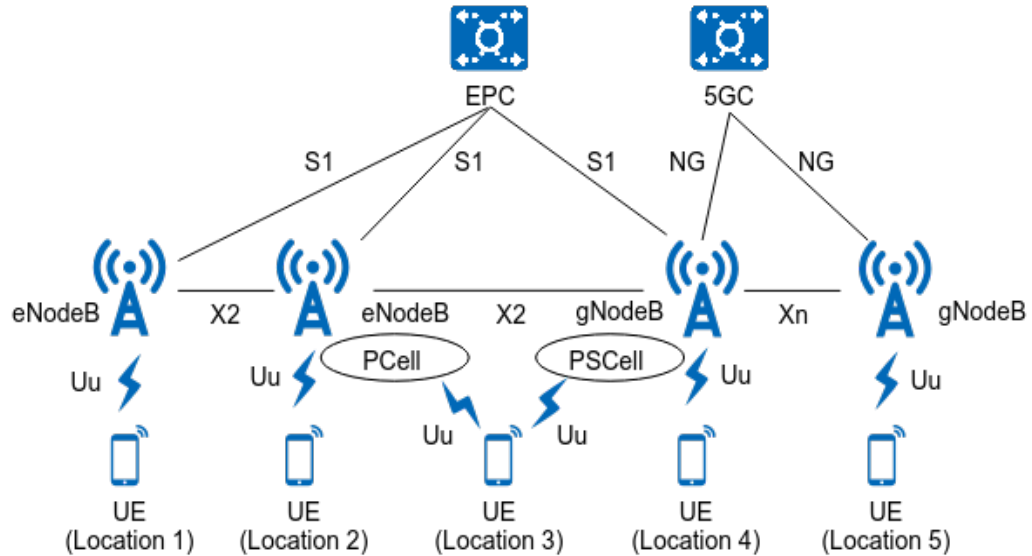
The RRC_INACTIVE state is not supported in NSA networking. Based on the RRC states of UEs, mobility management in NSA networking is categorized as follows:

- Mobility management in the RRC_CONNECTED state: UEs are connected to both the PCell and PSCell. Therefore, mobility management is categorized as PCell change and PSCell change. For details, see *NSA Mobility Management*.
- Mobility management in the RRC_IDLE state: UEs camp on LTE cells. For details, see *Idle Mode Management* in eRAN Feature Documentation.

NSA and SA Hybrid Mobility Management

NSA and SA hybrid networking marks a transition in the smooth evolution from NSA networking to SA networking, as shown in [Figure 3-4](#).

Figure 3-4 NSA and SA hybrid networking



As shown in [Figure 3-4](#), the following mobility scenarios occur in NSA and SA hybrid networking:

- When a UE moves from location 1 to location 2: The UE camps on an LTE cell at location 1 or location 2. In this scenario, intra-LTE mobility management is performed and the LTE mechanisms apply. For details, see *Idle Mode Management* and *Mobility Management in Connected Mode* in *eRAN Feature Documentation*.
- When a UE moves from location 2 to location 3: The UE camps on an LTE cell at location 2 and an NR cell in NSA networking at location 3. In this scenario, NSA mobility management is performed. For details, see *NSA Mobility Management*.
- When a UE moves from location 3 to location 4: The UE camps on an NR cell in NSA networking at location 3 and an NR cell in SA networking at location 4. In this scenario, interoperability between E-UTRAN and NG-RAN is performed. For details, see *NSA-SA Selection Based on User Experience*.
- When a UE moves from location 4 to location 5: The UE camps on an NR cell in SA networking at location 4 or location 5. In this scenario, SA mobility management is performed. If the UE is in connected mode during the movement, SA mobility management in connected mode is performed. For details, see *SA Mobility Management in Connected Mode*. If the UE is in idle mode during the movement, SA mobility management in idle mode is performed, as described in this document. If the UE is in inactive mode during the movement, SA mobility management in inactive mode is performed. For details, see *SA Mobility Management in Inactive Mode*.
- When a UE directly moves from location 1 to location 5: The UE camps on an LTE cell at location 1 and an NR cell in SA networking at location 5. In this scenario, interoperability between E-UTRAN and NG-RAN is performed. For

details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* and *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

4 Mobility Management in Idle Mode

Mobility management in idle mode typically refers to cell reselection. After camping on a cell, an RRC_IDLE UE measures the signal quality of the cell and that of neighboring cells, and reselects a higher-quality cell to camp on based on the cell reselection criteria.

This section describes the basic principles and mechanisms of mobility management in idle mode. Mobility management in idle mode mainly occurs on UEs and does not involve signaling exchanges between UEs and gNodeBs.

4.1 Principles

The following terms and definitions related to mobility management in idle mode apply:

- **Acceptable cell:** A cell that a UE obtains only limited services (such as emergency calls) from when camping on it.
A cell can be an acceptable cell only when:
 - It is not barred.
 - It meets the cell selection criteria (see [4.1.2 Cell Selection](#)).
- **Suitable cell:** A cell that a UE obtains normal services from when camping on it.
A cell can be a suitable cell only when:
 - It is not barred.
 - It meets the cell selection criteria (see [4.1.2 Cell Selection](#)).
 - It belongs to the UE-selected PLMN or a PLMN on the equivalent PLMN list.
 - It supports the PLMN selected or registered by the UE.
- **Barred cell:** A cell that is barred from providing services. If it is a single-operator cell, the master information block (MIB) is used to indicate its barred status. If it is a multi-operator cell, the SIB1 is used to indicate its barred status for each of the operators.

4.1.1 Cell Search and PLMN Selection

Cell Search

Cell search is a procedure for a UE to achieve time and frequency synchronization with a cell, obtain the physical cell identifier (PCI) of the cell, and learn the signal quality and other information about the cell based on the PCI.

In NR, two synchronization signals are used in cell search: primary synchronization signals (PSSs) and secondary synchronization signals (SSSs).

A UE performs cell search as follows:

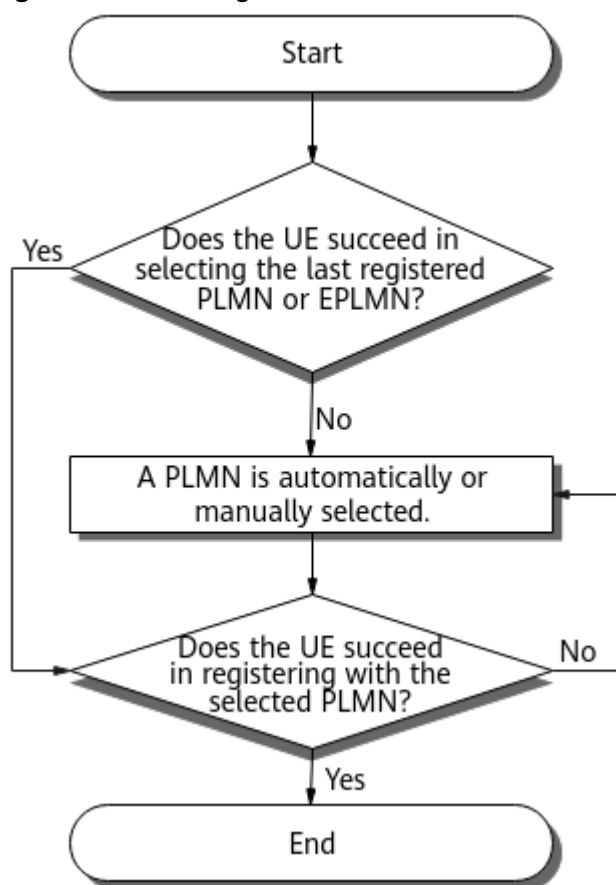
1. The UE detects the PSS to achieve clock synchronization and acquires an ID in a PCI group based on the mapping with the PSS.
2. The UE detects the SSS to achieve time synchronization (frame synchronization) and acquires a PCI group ID based on the mapping with the SSS.
3. The UE determines the cell's PCI based on the PCI group ID and the ID in the PCI group.
4. The UE detects the downlink SSB signal to determine the cell's signal quality.
5. The UE reads the MIB and SIB1 to obtain other information about the cell, such as the operator information.

PLMN Selection

After completing cell search, the UE selects and registers with a PLMN. After the PLMN registration succeeds, the UE displays the information about the PLMN and prepares to obtain services from the operator.

Figure 4-1 shows the procedure for the UE to select a PLMN.

Figure 4-1 Selecting a PLMN



4.1.2 Cell Selection

During a transition from RRC_CONNECTED to RRC_IDLE or after selecting a PLMN, the UE performs cell selection to choose a suitable cell to camp on.

A UE performs cell selection as follows:

1. The UE selects and camps on a suitable cell on the frequency indicated in the RRC Release message.
2. If a suitable cell is not found, the UE attempts to select the most recent cell it camped on while in connected mode as the suitable cell and camps on the cell.
3. If a suitable cell is still not found, the UE performs the Stored Information Cell Selection procedure to find and camp on a suitable cell.
4. If a suitable cell is still not found, the UE initiates the Initial Cell Selection procedure to find and camp on a suitable cell.
5. If the UE fails to find a suitable cell through the Initial Cell Selection procedure, it enters the Any Cell Selection State.

In the Any Cell Selection State, the UE attempts to find an acceptable cell to camp on. After finding an acceptable cell, the UE camps on the cell and obtains limited services from it while periodically performing the Initial Cell Selection procedure to search for a suitable cell. For details on the tasks UEs perform in the Any Cell Selection State, see section 5.2.7 "Any Cell Selection state" in 3GPP TS 38.304 V15.4.0.

 NOTE

In the preceding procedure, "a suitable cell not found" indicates that no corresponding cell meets the conditions to be a suitable cell (see 4.1 Principles). As a result, the UE cannot camp on the cell or acquire normal services.

The details of the Stored Information Cell Selection and Initial Cell Selection procedures are as follows:

- **Stored Information Cell Selection**
The UE performs cell selection based on information about the frequencies and cell parameters in the Stored Information (retrieved from previously received measurement configurations or detected system information), attempting to find a suitable cell. This information helps accelerate cell selection.
- **Initial Cell Selection**
The UE scans all frequencies in the supported RATs to search for a suitable cell. On each frequency, the UE searches for the cell with the strongest signal and camps on the first suitable cell that it finds.

Cell Selection Criteria

During cell selection, the UE determines whether a cell meets the cell selection criteria.

The cell selection criteria (referred to as criterion S) are as follows:

$S_{rxlev} > 0$ and $S_{qual} > 0$

In the cell selection criteria, S_{qual} must be greater than 0 only if the **SIB1_RSRQ_SW** option of the **NRDUCellSelConfig.SibOptionalleInd** parameter is selected; only S_{rxlev} must be greater than 0 in other cases.

 NOTE

For details about cell selection criteria, see section 5.2.3.2 "Cell Selection Criterion" in 3GPP TS 38.304 V15.4.0.

$S_{rxlev} = Q_{rxlevmeas} - (Q_{rxlevmin} + Q_{rxlevminoffset}) - P_{compensation} - Q_{offsettemp}$

Table 4-1 Variables in calculation of S_{rxlev}

Variabl e	Meaning	Parameter Configuratio n
S_{rxlev}	Cell selection RX level value (dB).	N/A
$Q_{rxlevmeas}$	Measured cell RX level value (RSRP).	N/A
$Q_{rxlevmin}$	Minimum required RX level (dBm) in the cell, which is broadcast in SIB1.	NRDUCellSel Config.Minim umRxLevel

Variable	Meaning	Parameter Configuration
$Q_{rxlevminoffset}$	Offset to the signaled $Q_{rxlevmin}$ (broadcast in SIB1). In the current version, this variable is not sent to UEs, which consider it to be 0 dB by default.	N/A
$P_{compensation}$	This variable equals the maximum value between ($P_{EMAX1} - P_{PowerClass}$) and 0. P_{EMAX1} is the SIB1-signalized maximum TX power of a UE that is allowed in a cell (indicated by the <i>P-Max</i> IE and specified by the NRDUCellSelConfig.MaxUeTransmitPower parameter) and $P_{PowerClass}$ is the UE's maximum RF output power.	N/A
$Q_{offsettemp}$	Offset (dB) temporarily applied to a cell and indicated by the <i>connEstFailOffset</i> sub-IE in the <i>ConnEstFailureControl</i> IE broadcast in SIB1. $Q_{offsettemp}$ indicates an optional calculation factor. In the current version, the value of $Q_{offsettemp}$ is always 1 dB. For details about the definition of $Q_{offsettemp}$, see section 5.3.3.7 "T300 expiry" in 3GPP TS 38.331. Assume that a UE fails to access a cell (the access failure indicated by 1) and the number of consecutive failures exceeds a specific threshold (indicated by 2). In this case, if the UE performs cell reselection evaluation before the expiration of a specific period (indicated by 3), $Q_{offsettemp}$ is used to calculate the signal quality ranking of the cell. - Expiration of timer T300, which is specified by NRDUCellUeTimerConst.T300 - The <i>ConnEstFailCount</i> sub-IE, which is fixed at 1 in the current version, in the <i>ConnEstFailureControl</i> IE broadcast in SIB1 in the serving cell - The <i>connEstFailOffsetValidity</i> sub-IE, which is fixed at 30s in the current version, in the <i>ConnEstFailureControl</i> IE broadcast in SIB1 in the serving cell	N/A

$$S_{qual} = Q_{qualmeas} - (Q_{qualmin} + Q_{qualminoffset}) - Q_{offsettemp}$$

Table 4-2 Variables in calculation of S_{qual}

Variable	Meaning	Parameter Configuration
S_{qual}	Cell selection quality value (dB).	N/A
$Q_{qualmeas}$	Measured cell quality value (RSRQ).	N/A
$Q_{qualmin}$	Minimum required quality level (dB) in the cell, which is broadcast in SIB1.	NRDUCellSel Config.Minim umRxQlty
$Q_{qualminoffset}$	Offset to the signaled $Q_{qualmin}$ (broadcast in SIB1). It is not signaled in the current version.	N/A
$Q_{offsettemp}$	Offset (dB) temporarily applied to a cell and indicated by the <i>connEstFailOffset</i> sub-IE in the <i>ConnEstFailureControl</i> IE broadcast in SIB1. $Q_{offsettemp}$ indicates an optional calculation factor. In the current version, the value of $Q_{offsettemp}$ is always 1 dB. For details about the definition of $Q_{offsettemp}$, see section 5.3.3.7 "T300 expiry" in 3GPP TS 38.331. Assume that a UE fails to access a cell (the access failure indicated by 1) and the number of consecutive failures exceeds a specific threshold (indicated by 2). In this case, if the UE performs cell reselection evaluation before the expiration of a specific period (indicated by 3), $Q_{offsettemp}$ is used to calculate the signal quality ranking of the cell. - Expiration of timer T300, which is specified by NRDUCellUeTimerConst. T300 - The <i>ConnEstFailCount</i> sub-IE, which is fixed at 1 in the current version, in the <i>ConnEstFailureControl</i> IE broadcast in SIB1 in the serving cell - The <i>connEstFailOffsetValidity</i> sub-IE, which is fixed at 30s in the current version, in the <i>ConnEstFailureControl</i> IE broadcast in SIB1 in the serving cell	N/A

4.1.3 Cell Reselection

After camping on a cell, the UE measures the signal quality of the cell and that of neighboring cells, and selects a higher-quality cell to camp on based on the cell reselection criteria.

For a UE to perform cell reselection, the gNodeB filters out the non-serving frequencies for which the non-serving frequency filtering function is enabled in the cell on which the UE camps. That is, the information about such frequencies is not

included in the RRCRelease message. The non-serving frequency filtering function is controlled by the **MOB_NEIGHBOR_FREQ_FILTER_SW** option of the **NRCellFreqRelation.FrequencyCtrlSwitch** parameter.

A UE performs cell reselection as follows:

1. Determines whether to start measuring neighboring cells based on the serving cell's S_{rxlev} and S_{qual} and the cell reselection priorities of the frequencies where the neighboring cells work. For details, see [4.1.3.1 Cell Reselection Priority](#) and [4.1.3.3 Neighboring Cell Measurement for Cell Reselection](#).
2. Performs cell reselection based on the measurement results of neighboring cells and the cell reselection criteria, which are related to the cell reselection priorities of the frequencies where the neighboring cells work.
 - For equal-priority frequencies, see [4.1.3.4 Intra-Frequency or Equal-Priority Inter-Frequency Cell Reselection](#).
 - For different-priority frequencies, see [4.1.3.5 Different-Priority Inter-Frequency Cell Reselection](#).

NOTE

In terms of the time when UEs are migrated and when the cell is deactivated, cell blocking can be categorized as high-priority blocking, medium-priority blocking, and low-priority blocking. To perform medium-priority blocking for a cell, the RRC_IDLE UEs in the cell must be migrated out. The gNodeB selects cell reselection for transferring UEs out of the cell.

4.1.3.1 Cell Reselection Priority

During measurements and cell reselections, UEs consider the cell reselection priorities of neighboring frequencies.

Cell reselection priorities are classified into common priorities and dedicated priorities. If both common and dedicated priorities are configured, dedicated priorities preferentially take effect.

Depending on service types, dedicated cell reselection priorities are further classified into the following types, which are delivered to UEs in the following sequence:

1. Cell reselection priorities dedicated to 3CC and massive CA UEs, as described in section "Reselection Policy for 3CC and Massive CA UEs in Idle Mode" in *CA Experience Improvement*
2. UE-layering-based differentiated cell reselection priorities, as described in [UE-Layering-based Differentiated Cell Reselection Priority Configuration](#)
3. RFSP-specific cell reselection priorities, as described in section "RFSP-based Cell Reselection Policy" in *Flexible User Steering*
4. QCI-specific cell reselection priorities, as described in section "Differentiated Carrier Selection for FWA Users" in *FWA*
5. Operator-specific cell reselection priorities, as described in section "Operator-Specific Dedicated Cell Reselection Priority Management" in *Multi-Operator Sharing*

4.1.3.1.1 Common Cell Reselection Priority

The common cell reselection priorities of different frequency types are described as follows:

- For the serving frequency, common cell reselection priorities are delivered to UEs through SIB2 broadcast and can be used by all UEs in the cell.
 - The cell reselection priority is configured by the **NRCellReselConfig.CellReselPriority** parameter. A larger value of this parameter indicates a higher priority.
 - The cell reselection sub-priority is configured by the **NRCellReselConfig.CellReselSubPriority** parameter. A larger value of this parameter indicates a higher priority. If this parameter is set to **ODOT0**, the cell reselection sub-priority is not delivered.

The UE adds up the cell reselection priority and sub-priority of a frequency and uses the sum as the final cell reselection priority for the frequency.

- For non-serving frequencies, common cell reselection priorities are delivered to UEs through SIB4 broadcast and can be used by all UEs in the cell.
 - The cell reselection priority is configured by the **NRCellFreqRelation.CellReselPriority** parameter. A larger value of this parameter indicates a higher priority. If this parameter is set to **255**, the cell reselection priority is not delivered.
 - The cell reselection sub-priority is configured by the **NRCellFreqRelation.CellReselSubPriority** parameter. A larger value of this parameter indicates a higher priority. If this parameter is set to **ODOT0**, the cell reselection sub-priority is not delivered.

The UE adds up the cell reselection priority and sub-priority of a frequency and uses the sum as the final cell reselection priority for the frequency.

For intra-frequency cell reselection, UEs ignore the priority information as the intra-frequency cells have the same cell reselection priority.

4.1.3.1.2 Dedicated Cell Reselection Priority

Dedicated cell reselection priorities and the T320 timer are delivered through the *CellReselectionPriorities* IE of the RRCRelease message during RRC connection release. The T320 timer specifies the validity period of dedicated cell reselection priorities. This timer is configured as follows:

- In RRC connection release triggered by load balancing, the T320 timer is configured by **NRCellReselConfig.T320ForMlb**. For details about the T320 timer in load balancing scenarios, see *Mobility Load Balancing in Idle Mode*.
- In RRC connection release not triggered by load balancing, the T320 timer is configured by **NRCellReselConfig.T320ForOther**.

The T320 timer starts upon reception of the *t320* IE on UEs. Before the T320 timer expires, UEs use the dedicated cell reselection priorities indicated in the current RRCRelease message for measurements and cell reselections. Once the T320 timer expires or UEs enter the RRC_CONNECTED state, these dedicated priorities are no longer valid. In that case, UEs use common cell reselection priorities for measurements and cell reselections. For details about the T320 timer, see section 7.1.1 "Timers (Informative)" in 3GPP TS 38.331.

To ensure differentiated user experience, **UE-Layering-based Differentiated Cell Reselection Priority Configuration** is introduced to support differentiated configuration of dedicated cell reselection priorities for UEs running services with different QCI.

UE-Layering-based Differentiated Cell Reselection Priority Configuration

UE-layering-based differentiated cell reselection priority configuration is designed to ensure differentiated user experience. This function associates a UE's QCI with a configured gNodeB frequency priority group ID (specified by the **gNBFreqPriorityGroup.gNBFreqPriorityGroupId** parameter) by using the **gNBDiffUserExpGroup** MO. The same QCI can be associated only with the same **gNBDiffUserExpGroup** MO.

- If **gNBDiffUserExpGroup.gNBDiffLayerParaGroupId** is set to a value other than **255**, the function takes effect. In this case, **gNBFreqPriorityGroup.CellReselPri** and **gNBFreqPriorityGroup.CellReselSubPri** for the gNodeB frequency priority group that is associated with **gNBDiffUserExpGroup.gNBDiffLayerParaGroupId** take effect.
- If **gNBDiffUserExpGroup.gNBDiffLayerParaGroupId** is set to **255**, the function does not take effect for the QCI that is associated with the **gNBDiffUserExpGroup** MO.

UE-layering-based differentiated cell reselection priority configuration can be disabled for all QCIs by selecting the **DIFF_UE_RESEL_PRI_SUPPR_SW** option of the **NRCellMobilityConfig.IdleReselPolicySw** parameter.

4.1.3.2 Cell Blacklisting for Cell Reselection

When sending system information to a UE, the serving cell includes the information about blacklisted intra-frequency cells in *IntraFreqExcludedCellList* of SIB3 and the information about blacklisted inter-frequency cells in *InterFreqExcludedCellList* of SIB4. The UE does not measure or reselect to any of the blacklisted cells broadcast in the received system information.

Configuration of a Cell Blacklist

In blacklisted cell configuration, a cell blacklist is configured at the base station level and then is associated with the serving cell.

1. The **gNBCellBlacklist** MO is used to configure a cell blacklist at the base station level.
 - a. A cell blacklist is identified by **gNBCellBlacklist.CellBlacklistId**, with **gNBCellBlacklist.RatType** set to **NR**.
 - b. A maximum of 16 blacklisted cell ranges can be configured in each cell blacklist. A blacklisted cell range is specified by the start physical cell ID and physical cell ID range. *N* consecutive PCI-specific cells starting from the one specified by the start physical cell ID are blacklisted.
where
 - The start physical cell ID is specified by **gNBCellBlackList.StartPhysicalCellId**.

- The physical cell ID range is specified by **gNBCellBlackList.PhysicalCellIdRange**.
 - N is a positive integer to which the value of **gNBCellBlackList.PhysicalCellIdRange** is mapped.
2. A cell blacklist is associated with the serving cell.
- Association with blacklisted intra-frequency cells
The intra-frequency cell blacklist identified by **gNBCellBlackList.CellBlacklistId** is associated with the serving cell through **NRCellMobilityConfig.IntraFExcludedCellListId**. This way, the blacklisted intra-frequency cells on the list identified by **gNBCellBlackList.CellBlacklistId** are associated with the serving cell.
 - Association with blacklisted inter-frequency cells
The inter-frequency cell blacklist identified by **gNBCellBlacklist.CellBlacklistId** is associated with the serving cell through **NRCellFreqRelation.InterFExcludedCellListId**. This way, the blacklisted inter-frequency cells on the list identified by **gNBCellBlacklist.CellBlacklistId** are associated with the serving cell.

NOTE

If **NRCellMobilityConfig.IntraFExcludedCellListId** is set to a value other than 65535, the value must be already configured in **gNBCellBlackList.CellBlacklistId** and the corresponding **gNBCellBlacklist.RatType** value must be NR.

If **NRCellFreqRelation.InterFExcludedCellListId** is set to a value other than 65535, the value must be already configured in **gNBCellBlackList.CellBlacklistId** and the corresponding **gNBCellBlacklist.RatType** value must be NR.

The values of **gNBCellBlacklist.RatType** in the records that have the same value of **gNBCellBlacklist.CellBlacklistId** must also be the same.

Cell Blacklist Delivery Specifications

SIB4 carries the configuration information about a maximum of eight frequencies, with a maximum of 16 blacklisted cell ranges configured for each frequency. Due to the limited length of SIB4, a total of up to 16 blacklisted cell ranges for all non-serving frequencies can be carried in SIB4.

If the total number of blacklisted cell ranges for all non-serving frequencies exceeds 16, the 16 blacklisted cell ranges actually carried in SIB4 are selected according to the following principles:

- Blacklisted cell ranges for high-priority frequencies are preferentially selected. If multiple frequencies have the same priority, the blacklisted cell ranges for NR frequencies with smaller values of *SsbFreqPos* are preferentially selected. For details about frequency priorities, see [4.1.3.1 Cell Reselection Priority](#).
- If not all blacklisted cell ranges for a single frequency can be selected, the ones with larger values of **gNBCellBlackList.PhysicalCellIdRange** are preferentially selected. If multiple blacklisted cell ranges have the same value of **gNBCellBlackList.PhysicalCellIdRange**, the ones with smaller values of **gNBCellBlackList.StartPhysicalCellId** are preferentially selected.

4.1.3.3 Neighboring Cell Measurement for Cell Reselection

Based on the S_{rxlev} and S_{qual} of the serving cell and the cell reselection priorities of neighboring frequencies, a UE determines whether to measure the frequencies broadcast in the system information.

- Intra-frequency measurement
 - If the S_{rxlev} of the serving cell is greater than the value of the **NRCellReselConfig.IntraFreqMeasStartThld** parameter, the UE does not start neighboring cell measurements for intra-frequency cell reselection.
 - If the S_{rxlev} of the serving cell is less than or equal to the value of the **NRCellReselConfig.IntraFreqMeasStartThld** parameter, the UE starts neighboring cell measurements for intra-frequency cell reselection.
- Inter-frequency measurement
 - If a non-serving frequency has a higher priority than the serving frequency, the UE starts neighboring cell measurements for cell reselection regardless of the serving cell's signal quality.
 - If a non-serving frequency has an equal or lower priority than the serving frequency:
 - If the S_{rxlev} of the serving cell is greater than the value of **NRCellReselConfig.NonIntraFreqMeasRsrpThld** and the S_{qual} is greater than the value of **NRCellReselConfig.NonIntraFreqMeasRsrqThld**, the UE does not start neighboring cell measurements for inter-frequency cell reselection.
 - If the S_{rxlev} of the serving cell is less than or equal to the value of **NRCellReselConfig.NonIntraFreqMeasRsrpThld** or the S_{qual} is less than or equal to the value of **NRCellReselConfig.NonIntraFreqMeasRsrqThld**, the UE starts neighboring cell measurements for inter-frequency cell reselection.

SMTC Measurement Configuration

5G cell reselections are based on synchronization signal and PBCH block (SSB) measurements performed by UEs. Measurement of all SSBs in a cell is preferred for a more accurate SSB measurement result. A cell sends SSBs in periodic scanning mode, and SSBs are not sent in all time-domain positions within a scanning period. If UEs search for and measure SSBs in all time-domain positions, a large amount of power is wasted. **To effectively indicate the time window for UEs to measure SSBs and reduce unnecessary measurement power consumption, 5G introduces SSB-based measurement timing configuration (SMTC), which is a time window configured by the gNodeB for UEs to measure SSBs. UEs measure SSBs only within the time window.**

SMTC includes an SMTC period, SMTC duration, and SMTC offset. SMTC for different frequency types is delivered through the following messages:

- For the serving frequency, SMTC is delivered to UEs through SIB2 broadcast and can be used by all UEs in the cell.
- For non-serving frequencies, SMTC is delivered to UEs through SIB4 broadcast and can be used by all UEs in the cell.

For details about SMTC, see *SA Mobility Management in Connected Mode*.

Configuration of SSB Beam Measurement Result Selection and Combination

A UE needs to combine the SSB beam-level measurement results of a cell when calculating the cell-level SSB measurement result. These beam-level results must meet the SSB combination threshold requirement.

- Intra-frequency cell measurement: The SSB combination threshold is specified by the **NRCelReselConfig.SsbConsolidationThld** parameter. Upon detecting that multiple SSB beams in a cell meet the combination threshold requirement, the UE combines the measurement results of these SSB beams into a cell-level result. The maximum number of SSB beams whose measurement results can be combined is specified by the **NRCelReselConfig.SsbNumToAverage** parameter.
- Inter-frequency cell measurement: The SSB combination threshold is specified by the **NRCelMobilityConfig.SsbConsolidationThld** parameter. Upon detecting that multiple SSB beams in a cell meet the combination threshold requirement, the UE combines the measurement results of these SSB beams into a cell-level result. The maximum number of SSB beams whose measurement results can be combined is specified by the **NRCelMobilityConfig.SsbNumToAverage** parameter.

Optimized Configuration for deriveSSB-IndexFromCell

The gNodeB includes the *deriveSSB-IndexFromCell* IE (either set to **TRUE** or **FALSE**) in SIB2 or SIB4. For intra-frequency neighboring cell measurements, the value of this IE indicates whether UEs derive the SSB indexes of all intra-frequency neighboring cells based on the timing information of the serving cell. For inter-frequency neighboring cell measurements, it indicates whether UEs derive the SSB indexes of all neighboring cells on the frequency of any detected inter-frequency cell based on the timing information of that cell. If all cells on a frequency are time-synchronized, you can improve the efficiency and accuracy of detecting neighboring cells' SSB indexes by setting this IE to **TRUE**.

Optimized configuration for deriveSSB-IndexFromCell has been introduced. This function is enabled if the **PROTOCOL_IE_OPT_SW** option the **NRCelAlgoSwitch.HoCompatibilitySwitch** parameter is selected. When this function is enabled, the value of the *deriveSSB-IndexFromCell* IE in SIB2 or SIB4 sent from the gNodeB varies with the RATs corresponding to the frequencies that are to be measured. If an NR TDD frequency is to be measured, the value of the *deriveSSB-IndexFromCell* IE is **TRUE**.

4.1.3.4 Intra-Frequency or Equal-Priority Inter-Frequency Cell Reselection

After measuring neighboring cells for reselection, the UE performs intra-frequency or equal-priority inter-frequency cell reselection as follows:

1. The UE selects the neighboring cells that meet the cell selection criteria (criterion S, which is described in [Cell Selection Criteria](#)). The parameters related to intra- and inter-frequency neighboring cells are as follows:
In the cell selection criteria, S_{qual} must be greater than 0 only if the **RSRQ_BASED_RESEL_SW** option of the **NRCelReselConfig.SibOptionalleInd** parameter is selected; only S_{rxlev} must be greater than 0 in other cases.

- The S_{rxlev} of an intra-frequency neighboring cell is calculated using the $Q_{rxlevmin}$ that is equal to the minimum cell receive level (configured by the **NRCellReselConfig.MinimumRxLevel** parameter) broadcast in SIB2.
 - The S_{rxlev} of an inter-frequency neighboring cell is calculated using the $Q_{rxlevmin}$ that is equal to the minimum cell receive level (configured by the **NRCellFreqRelation.MinimumRxLevel** parameter) broadcast in SIB4.
 - The S_{qual} of an inter-frequency neighboring cell is calculated using the $Q_{qualmin}$ that is equal to the minimum required quality level in the cell (configured by the **NRCellFreqRelation.MinimumRxQlty** parameter) broadcast in SIB4.
2. Among the serving cell and the neighboring cells that meet the cell selection criteria, the UE selects the cell with the highest signal quality ranking as the highest ranked cell.

The signal quality ranking of a cell is calculated as follows:

- **Signal quality ranking of a neighboring cell:** $R_n = Q_{meas,n} - Q_{offset} - Q_{offsettemp}$
- **Signal quality ranking of the serving cell:** $R_s = Q_{meas,s} + Q_{hyst} - Q_{offsettemp}$

Table 4-3 describes the variables involved in the above formulas.

Table 4-3 Variables in calculation of the signal quality ranking of a cell

Variable	Meaning	Parameter Configuration
$Q_{meas,n}$	SSB-based measured receive level, which is also known as RSRP, of a neighboring cell.	N/A
Q_{offset}	Offset for reselection to a neighboring cell. The Q_{offset} of an intra-frequency neighboring cell is equal to q-OffsetCell broadcast in SIB3. Q_{offset} of an inter-frequency neighboring cell is equal to the sum of q-OffsetCell and q-OffsetFreq broadcast in SIB4.	• q-OffsetCell: specified by NRCellRelation.NCellReselOffset • q-OffsetFreq: specified by NRCellFreqRelation.FreqOffset
$Q_{meas,s}$	SSB-based measured receive level, which is also known as RSRP, of the serving cell.	N/A
Q_{hyst}	Hysteresis for cell reselection.	NRCellReselConfig.CellReselHysteresis

Variable	Meaning	Parameter Configuration
$Q_{\text{offsettemp}}$	$Q_{\text{offsettemp}}$ is the offset (dB) temporarily applied to a cell and indicated by the <i>connEstFailOffset</i> sub-IE in the <i>ConnEstFailureControl</i> IE broadcast in SIB1 in the serving cell. $Q_{\text{offsettemp}}$ indicates an optional calculation factor. In the current version, the value of $Q_{\text{offsettemp}}$ is always 1 dB. For details about the definition of $Q_{\text{offsettemp}}$, see section 5.3.3.7 "T300 expiry" in 3GPP TS 38.331. Assume that a UE fails to access a neighboring cell (the access failure is indicated by 1) and the number of consecutive failures exceeds a specific threshold (indicated by 2). In this case, if the UE performs cell reselection evaluation before the expiration of the specific period (indicated by 3), $Q_{\text{offsettemp}}$ is used to calculate the signal quality ranking of the neighboring cell only, but not of other cells. - Expiration of timer T300, which is specified by NRDUCellUeTimerConst. T300 - The <i>ConnEstFailCount</i> sub-IE, which is fixed at 1 in the current version, in the <i>ConnEstFailureControl</i> IE broadcast in SIB1 in the serving cell - The <i>connEstFailOffsetValidity</i> sub-IE, which is fixed at 30s in the current version, in the <i>ConnEstFailureControl</i> IE broadcast in SIB1 in the serving cell	N/A

- Among the serving cell and the neighboring cells that meet the cell selection criteria, the UE identifies the cells whose signal quality rankings meet the following condition:

$$R_{\text{highest ranked cell}} - R \leq \text{rangeToBestCell}$$

In the above formula:

- $R_{\text{highest ranked cell}}$ is the signal quality ranking of the highest ranked cell. For calculation details, see 2.
 - R is the signal quality ranking of the serving cell or a neighboring cell that meets the cell selection criteria. For calculation details, see 2.
 - rangeToBestCell is the threshold for the signal quality rankings of candidate cells, and is specified by the **NRCellReselConfig.CandidateCellThld** parameter. rangeToBestCell is indicated in SIB2.
4. Among the highest ranked cell and the cells that meet the preceding condition, the UE selects the cell with the maximum number of beams whose beam-level RSRP is greater than the SSB combination threshold as the best cell.
- If there are multiple such cells, the UE selects the cell with the highest signal quality ranking as the best cell. If none of the cells has a beam whose beam-level RSRP is greater than the SSB combination threshold, the UE directly selects the cell with the highest signal quality ranking among the highest ranked cell and the cells that meet the preceding condition as the best cell.
- For intra-frequency neighboring cells, the SSB combination threshold is specified by the **NRCellReselConfig.SsbConsolidationThld** parameter and sent to the UE through the *absThreshSS-BlocksConsolidation* IE in SIB2.
 - For inter-frequency neighboring cells, the SSB combination threshold is specified by the **NRCellMobilityConfig.SsbConsolidationThld** parameter and sent to the UE through the *absThreshSS-BlocksConsolidation* IE in SIB4.
5. If the cell selected as the best cell is not the serving cell, the UE checks whether the best cell meets specific conditions. If it does, the UE reselects to the cell. Otherwise, the UE continues to camp on the original cell. The specific conditions are as follows:
- The best cell meets the preceding cell reselection criteria (described in steps 1 through 4) throughout the time specified by $T_{\text{reselection}_{\text{NR}}}$ (fixed at 1s).
 - The UE has camped on the serving cell for more than 1s.

4.1.3.5 Different-Priority Inter-Frequency Cell Reselection

After measuring neighboring cells for reselection, the UE first selects inter-frequency neighboring cells that meet the cell selection criteria (criterion S, which is described in [Cell Selection Criteria](#)) if different priorities are configured for the frequencies of inter-frequency neighboring cells. In the cell selection criteria, S_{qual} must be greater than 0 only if the **RSRQ_BASED_RESEL_SW** option of the **NRCellReselConfig.SibOptionalleInd** parameter is selected; only S_{rxlev} must be greater than 0 in other cases.

- The S_{rxlev} of an inter-frequency neighboring cell is calculated using the Q_{rxlevmin} that is equal to the minimum cell receive level (configured by the **NRCellFreqRelation.MinimumRxLevel** parameter) broadcast in SIB4.
- The S_{qual} of an inter-frequency neighboring cell is calculated using the Q_{qualmin} that is equal to the minimum required quality level in the cell

(configured by the **NRCellFreqRelation.MinimumRxQlty** parameter)
broadcast in SIB4.

Then, the UE further selects the target cell for cell reselection among the inter-frequency neighboring cells that meet the cell selection criteria, based on the different-priority inter-frequency cell reselection criteria. With different-priority inter-frequency cell reselection, the UE can reselect to the cells on frequencies with higher or lower cell reselection priorities. For details about cell reselection priority configuration, see [4.1.3.1 Cell Reselection Priority](#).

- If all of the following conditions are met, the UE reselects to a cell on a non-serving frequency with a higher priority:
 - Condition 1: The UE has camped on the serving cell for more than 1s.
 - Condition 2:
 - If the **RSRQ_BASED_RESEL_SW** option of the **NRCellReselConfig.SibOptionalleInd** parameter is deselected, the S_{rxlev} of the neighboring cell that is evaluated is greater than *threshX-HighP* broadcast in SIB4 throughout the period specified by $T_{reselection_{NR}}$ (fixed at 1s) broadcast in SIB4.
 - If the **RSRQ_BASED_RESEL_SW** option of the **NRCellReselConfig.SibOptionalleInd** parameter is selected, the S_{qual} of the neighboring cell that is evaluated is greater than *threshX-HighQ* broadcast in SIB4 throughout the period specified by $T_{reselection_{NR}}$ (fixed at 1s) broadcast in SIB4.

If multiple cells meet the preceding conditions, the cell with the highest priority is selected for cell reselection. If there are still multiple cells selected, the neighboring cell with the highest R_n (signal quality ranking, as determined in 2 of [4.1.3.4 Intra-Frequency or Equal-Priority Inter-Frequency Cell Reselection](#)) is selected for cell reselection.

threshX-HighP and *threshX-HighQ* can be specified by the following parameters:

- *threshX-HighP*: specified by **NRCellFreqRelation.HighPriReselThld**
- *threshX-HighQ*: specified by **NRCellFreqRelation.HighPriReselRsrqThld**
- If all of the following conditions are met, the UE reselects to a cell on a non-serving frequency with a lower priority:
 - Condition 1: No neighboring cells on non-serving frequencies with a higher priority meet the criteria for reselecting to a cell on a frequency with a higher priority.
 - Condition 2: The UE has camped on the serving cell for more than 1s.
 - Condition 3:
 - If the **RSRQ_BASED_RESEL_SW** option of the **NRCellReselConfig.SibOptionalleInd** parameter is deselected and SIB2 does not include *threshServingLowQ*, the following constraints must be satisfied throughout the period specified by $T_{reselection_{NR}}$ (fixed at 1s) broadcast in SIB4:
 - The S_{rxlev} of the serving cell is less than *threshServingLowP* broadcast in SIB2.

- The S_{rxlev} of the neighboring cell that is evaluated is greater than *threshX-LowP* broadcast in SIB4.
- If the **RSRQ_BASED_RESEL_SW** option of the **NRCellReselConfig.SibOptionalleInd** parameter is selected and SIB2 includes *threshServingLowQ*, the following constraints must be satisfied throughout the period specified by *Treselection_{NR}* (fixed at 1s) broadcast in SIB4:
 - The S_{qual} of the serving cell is less than *threshServingLowQ* broadcast in SIB2.
 - The S_{qual} of the neighboring cell that is evaluated is greater than *threshX-LowQ* broadcast in SIB4.

If multiple cells meet the preceding conditions, the cell with the highest priority is selected for cell reselection. If there are still multiple cells selected, the neighboring cell with the highest R_n (signal quality ranking, as determined in 2 of 4.1.3.4 Intra-Frequency or Equal-Priority Inter-Frequency Cell Reselection) is selected for cell reselection.

threshServingLowP, *threshX-LowP*, *threshServingLowQ*, and *threshX-LowQ* can be specified by the following parameters:

- *threshServingLowP*: specified by **NRCellReselConfig.ServFreqLowPriRsrpReselThd**
- *threshX-LowP*: specified by **NRCellFreqRelation.LowPriReselThld**
- *threshServingLowQ*: specified by **NRCellReselConfig.ServFreqLowPriRsrqReselThd**
- *threshX-LowQ*: specified by **NRCellFreqRelation.LowPriReselRsrqThld**

4.1.3.6 Speed-based Cell Reselection

Moving UEs are classified, by speed, into normal-mobility UEs, medium-mobility UEs, and high-mobility UEs. A UE may also start at a normal speed and then accelerate. When the UE is moving at a medium or high speed, it reselects a cell quickly and its cell reselection parameters need to be adjusted accordingly.

In this context, speed-based cell reselection is introduced. This function is supported only in low frequency bands. This function is controlled by the **SPEED_BASED_RESEL_SW** option of the **NRCellReselConfig.SibOptionalleInd** parameter.

To enable speed-based cell reselection to inter-frequency or inter-RAT cells, further select the **INTER_FREQ_RAT_SPEED_RESEL_SW** option of the **NRCellReselConfig.SibOptionalleInd** parameter. This requirement does not apply to speed-based cell reselection to intra-frequency cells. Note that the **SPEED_BASED_RESEL_SW** option of the **NRCellReselConfig.SibOptionalleInd** parameter must be selected before you select the **INTER_FREQ_RAT_SPEED_RESEL_SW** option of the **NRCellReselConfig.SibOptionalleInd** parameter.

When speed-based cell reselection is enabled, a UE can evaluate its speed based on the system information (SI) broadcast by the gNodeB and then use appropriate cell reselection parameters adaptive to its speed for cell reselection.

- A UE evaluates its speed based on the SI broadcast by the gNodeB. The criteria for evaluation are as follows:
 - Normal-mobility UE
The number of reselected cells is less than or equal to $n\text{-CellChangeMedium}$ within a $t\text{-Evaluation}$ duration and the number of reselected cells is less than or equal to $n\text{-CellChangeMedium}$ within the subsequent $t\text{-HystNormal}$ duration (fixed at 30s).
 - Medium-mobility UE
The number of reselected cells is greater than $n\text{-CellChangeMedium}$ but less than or equal to $n\text{-CellChangeHigh}$ within a $t\text{-Evaluation}$ duration.
 - High-mobility UE
The number of reselected cells is greater than $n\text{-CellChangeHigh}$ within a $t\text{-Evaluation}$ duration.

where

- $t\text{-Evaluation}$ is the UE mobility state evaluation period, which is specified by the **NRCellReselConfig.MobStateEvalPeriod** parameter.
- $n\text{-CellChangeMedium}$ is the threshold expressed as a number of cell reselections for evaluating whether UEs enter the medium-mobility state, which is specified by the **NRCellReselConfig.ReselThldForMediumMob** parameter.
- $n\text{-CellChangeHigh}$ is the threshold expressed as a number of cell reselections for evaluating whether UEs enter the high-mobility state, which is specified by the **NRCellReselConfig.ReselThldForHighMob** parameter.
- A UE selects appropriate cell reselection parameters adaptive to its speed for cell reselection.
 - Normal-mobility UE
Cell reselection is implemented according to the cell reselection parameters described in [4.1.3.4 Intra-Frequency or Equal-Priority Inter-Frequency Cell Reselection](#) or [4.1.3.5 Different-Priority Inter-Frequency Cell Reselection](#).
 - Medium-mobility UE
The cell reselection parameters described in [4.1.3.4 Intra-Frequency or Equal-Priority Inter-Frequency Cell Reselection](#) or [4.1.3.5 Different-Priority Inter-Frequency Cell Reselection](#) are modified as follows:
Calculation of the reselection time in the cell reselection principle (described in [4.1.3.4 Intra-Frequency or Equal-Priority Inter-Frequency Cell Reselection](#) or [4.1.3.5 Different-Priority Inter-Frequency Cell Reselection](#)): $\text{Reselection time} = T_{\text{reselection_NR}} \times (sf\text{-Medium IE in the SpeedStateScaleFactors IE})$
Calculation of the reselection hysteresis used to calculate the signal quality level of the current serving cell in the cell reselection principle (described in [4.1.3.4 Intra-Frequency or Equal-Priority Inter-Frequency Cell Reselection](#)): $\text{Reselection hysteresis} = Q_{\text{hyst}} + (sf\text{-Medium IE in the } q\text{-HystSF IE})$
where

- $T_{\text{reselection}_{\text{NR}}}$ is fixed at 1s.
 - Q_{hyst} is the hysteresis for cell reselection, which is configured by the **NRCellReselConfig.CellReselHysteresis** parameter.
 - *sf-Medium* IE in the *SpeedStateScaleFactors* IE is the scaling factor for the cell reselection time of medium-mobility UEs, which is specified by the **NRCellReselConfig.ReselTimeSfForMediumMob** parameter.
 - *sf-Medium* IE in the *q-HystSF* IE is the additional hysteresis for cell reselection by medium-mobility UEs, which is specified by the **NRCellReselConfig.AddlHystForMediumMob** parameter.
- High-mobility UE
- The cell reselection parameters described in [4.1.3.4 Intra-Frequency or Equal-Priority Inter-Frequency Cell Reselection](#) or [4.1.3.5 Different-Priority Inter-Frequency Cell Reselection](#) are modified as follows:
- Calculation of the reselection time in the cell reselection principle (described in [4.1.3.4 Intra-Frequency or Equal-Priority Inter-Frequency Cell Reselection](#) or [4.1.3.5 Different-Priority Inter-Frequency Cell Reselection](#)): $\text{Reselection time} = T_{\text{reselection}_{\text{NR}}} \times (\text{sf-High IE in the SpeedStateScaleFactors IE})$
- Calculation of the reselection hysteresis used to calculate the signal quality level of the current serving cell in the cell reselection principle (described in [4.1.3.4 Intra-Frequency or Equal-Priority Inter-Frequency Cell Reselection](#)): $\text{Reselection hysteresis} = Q_{\text{hyst}} + (\text{sf-High IE in the q-HystSF IE})$
- where
- $T_{\text{reselection}_{\text{NR}}}$ is fixed at 1s.
 - Q_{hyst} is the hysteresis for cell reselection, which is configured by the **NRCellReselConfig.CellReselHysteresis** parameter.
 - *sf-High* IE in the *SpeedStateScaleFactors* IE is the scaling factor for the cell reselection time of high-mobility UEs, which is specified by the **NRCellReselConfig.ReselTimeSfForHighMob** parameter.
 - *sf-High* IE in the *q-HystSF* IE is the additional hysteresis for cell reselection by high-mobility UEs, which is specified by the **NRCellReselConfig.AddlHystForHighMob** parameter.

4.2 Network Analysis

4.2.1 Benefits

This function allows UEs in RRC_IDLE mode to camp on a cell with better signal quality, ensuring continuous coverage and providing users with seamless service experience.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

- Non-serving frequency filtering

After this function is enabled for a non-serving frequency of the serving cell, the number of cell reselections initiated by UEs that camp on the serving cell to the cell operating on the non-serving frequency decreases.

- Timer T320

A smaller value of the timer results in a shorter duration before the cell-reselection parameters contained in the RRCRelease message become invalid. A larger value of the timer results in a longer duration, which may lead to delayed update of cell-reselection parameters on the UE side.

- UE-layering-based differentiated cell reselection priority configuration

This function may have the following impacts:

- If UEs are transferred to cells with higher cell reselection priorities, the uplink and downlink CCE allocation failure rates, **User Downlink Average Throughput (DU)**, **User Uplink Average Throughput (DU)**, **N.ThpVol.UL.Cell**, **N.ThpVol.DL.Cell**, and uplink and downlink PRB usage of the original cells decrease.

where

- $$\text{Uplink CCE allocation failure rate} = (\text{N.CCE.UL.AllocReq.Num} - \text{N.CCE.UL.AggLvl1Num} - \text{N.CCE.UL.AggLvl2Num} - \text{N.CCE.UL.AggLvl4Num} - \text{N.CCE.UL.AggLvl8Num} - \text{N.CCE.UL.AggLvl16Num}) / \text{N.CCE.UL.AllocReq.Num}$$
- $$\text{Downlink CCE allocation failure rate} = (\text{N.CCE.DL.AllocReq.Num} - \text{N.CCE.DL.AggLvl1Num} - \text{N.CCE.DL.AggLvl2Num} - \text{N.CCE.DL.AggLvl4Num} - \text{N.CCE.DL.AggLvl8Num} - \text{N.CCE.DL.AggLvl16Num}) / \text{N.CCE.DL.AllocReq.Num}$$
- $$\text{Uplink PRB usage} = (\text{N.PRB.UL.Used.Avg} / \text{N.PRB.UL.Avail.Avg}) \times 100\%$$
- $$\text{Downlink PRB usage} = (\text{N.PRB.DL.Used.Avg} / \text{N.PRB.DL.Avail.Avg}) \times 100\%$$
- Some UEs cannot access specific frequencies or cells. As a result, the **Service Call Drop Rate (CU)** or **Service Call Drop Rate (CU, Inactive)** increases.

4.3 Requirements

4.3.1 Licenses

There are no license requirements.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-frequency handover policy for voice-service-capable UEs	VOICE_INTER_FREQ_HO_POLICY_SWITCH option of the NRCellFreqRelationAggregationAttribute parameter	<i>SA Mobility Management in Connected Mode</i>	If the "inter-frequency handover policy for voice-service-capable UEs" is enabled for a visited network's frequency in a home network's cell, the home network's gNodeB will not send an RRCRelease message containing the frequency to a VoNR-incapable UE of the home operator in roaming scenarios, and the UE will not reselect this frequency based on dedicated cell reselection priorities.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	If the "voice-capability-based inter-frequency handover back to the HPLMN" function is enabled for the home network on the visited network's gNodeB, the gNodeB will send an RRCRelease message to a UE that does not support VoNR on the home network in roaming scenarios. The message contains the frequencies for which gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag is set to TRUE in the frequency list group bound to the UE on the gNodeB side. In this case, the UE can reselect the corresponding frequency based on dedicated cell reselection priorities.
Low-frequency TDD	Reselection policy for 3CC and massive CA UEs in idle mode	MASS_CA_IDLE_RESEL_POLICY_SW option of the NRCellOpQciBearer.ServiceDiffAlgoSwitch parameter	<i>CA Experience Improvement</i>	If both dedicated-priority-based idle mode MLB and the reselection policy for 3CC and massive CA UEs in idle mode are enabled, the T320 timer takes the value of NRCellReselConfig.T320ForOther for UEs in the 3CC aggregation state and massive CA UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dedicated-priority-based idle mode MLB	All of the following settings apply: NRCellMLb.IdleMLbPolicy set to DEDICATED_PRIORITY INTER_FREQ_IDLE_MLB_SW option of the NRCellAlgorithmSwitch.MlbAlgoSwitch parameter selected	<i>Mobility Load Balancing in Idle Mode</i>	

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

Speed-based cell reselection: UEs must support speed-based cell reselection.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

This function is enabled by default and does not need activation.

Table 4-4 describes the parameters used for function optimization.

Table 4-4 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Cell Reselection Hysteresis	NRCellReselConfig.<i>CellReselHysteresis</i>	Use the recommended value.
Intra-Freq Measurement Start Threshold	NRCellReselConfig.<i>IntraFreqMeasStartThld</i>	Use the recommended value.
Minimum RX Level	NRCellReselConfig.<i>MinimumRxLevel</i>	Use the recommended value.
Cell Reselection Priority	NRCellReselConfig.<i>CellReselPriority</i>	Use the recommended value.
Cell Reselection Sub-Priority	NRCellReselConfig.<i>CellReselSubPriority</i>	Use the recommended value.
SSB Consolidation Threshold	NRCellReselConfig.<i>SsbConsolidationThld</i>	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Candidate Cell Thld	NRCellReselConfig.CandidateCellThld	A smaller value of this parameter results in a smaller number of candidate cells that meet the threshold for the signal quality rankings and a lower probability of triggering cell reselection. A larger value of this parameter results in a larger number of candidate cells that meet the threshold for the signal quality rankings and a higher probability of triggering cell reselection. Use the recommended value.
SSB Number To Average	NRCellReselConfig.SsbNumToAverage	Use the recommended value.
Handover Compatibility Switch	NRCellAlgoSwitch.HoCompatibility-Switch	It is recommended that the PROTOCOL_IE_OPT_SW option be selected.
Non-Intra-Freq Measurement RSRP Threshold	NRCellReselConfig.NonIntraFreqMeasRsrpThld	Use the recommended value.
Non-Intra-Freq Measurement RSRQ Threshold	NRCellReselConfig.NonIntraFreqMeasRsrqThld	Use the recommended value.
Serving Freq Low Priority RSRP Resel Thld	NRCellReselConfig.ServFreqLowPriorityRsrpReselThld	Use the recommended value.
Minimum RX Level	NRDUCellSelConfig.MinimumRxLevel	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
SIB Optional IE Indicator	NRDUCellSelConfig.SibOptionalleInd	Select the SIB1_RSRQ_SW option if RSRQ-based cell selection is required.
Minimum RX Quality	NRDUCellSelConfig.MinimumRxQlty	Use the recommended value.
Neighboring Cell Reselection Offset	NRCellRelation.NCellReselOffset	Use the recommended value.
Frequency Offset	NRCellFreqRelation.FreqOffset	Use the recommended value.
Cell Reselection Priority	NRCellFreqRelation.CellReselPriority	Use the recommended value. Set this parameter to 255 for a non-serving frequency that is not used for cell reselection. In this way, the gNodeB does not deliver the cell reselection priority of the non-serving frequency to UEs.
Cell Reselection Sub-Priority	NRCellFreqRelation.CellReselSubPriority	Use the recommended value.
Cell Blacklist ID	gNBCellBlacklist.CellBlacklistId	Set this parameter based on the network plan.
Start Physical Cell ID	gNBCellBlacklist.StartPhysicalCellId	Set this parameter based on the network plan.
Physical Cell ID Range	gNBCellBlacklist.PhysicalCellIdRange	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
RAT Type	gNBCellBlacklist.RatType	Set this parameter based on the network plan. The values of gNBCellBlacklist.RatType in the records that have the same value of gNBCellBlacklist.CellBlacklistId must also be the same.
Intra-Frequency Excluded Cell List ID	NRCellMobilityConfig.IntraFExcludedCellListId	Set this parameter based on the network plan. If NRCellMobilityConfig.IntraFExcludedCellListId is set to a value other than 65535 , the value must have been configured in gNBCellBlackList.CellBlacklistId and the corresponding gNBCellBlacklist.RatType value must be NR .
Inter-Frequency Excluded Cell List ID	NRCellFreqRelation.InterFExcludedCellListId	Set this parameter based on the network plan. If NRCellFreqRelation.InterFExcludedCellListId is set to a value other than 65535 , the value must have been configured in gNBCellBlackList.CellBlacklistId and the corresponding gNBCellBlacklist.RatType value must be NR .

Parameter Name	Parameter ID	Setting Notes
SSB Consolidation Threshold	NRCellMobilityCon- fig.<i>SsbConsolidationThld</i>	Use the recommended value.
SSB Number To Average	NRCellMobilityCon- fig.<i>SsbNumToAverage</i>	Use the recommended value.
Minimum RX Level	NRCellFreqRelation.<i>MinimumRxLevel</i>	Use the recommended value.
Minimum RX Quality	NRCellFreqRelation.<i>MinimumRxQlty</i>	Use the recommended value.
High Priority Reselection Threshold	NRCellFreqRelation.<i>HighPriReselThld</i>	Use the recommended value.
SIB Optional IE Indicator	NRCellReselConfig.<i>SibOptionalleInd</i>	Select the RSRQ_BASED_RESEL_SW option if RSRQ-based cell reselection is required. Select the SPEED_BASED_RESEL_SW option if speed-based cell reselection is required. If speed-based cell reselection to inter-frequency or inter-RAT cells is required, further select the INTER_FREQ_RAT_SPEED_RESEL_SW option of the NRCellReselConfig.<i>SibOptionalleInd</i> parameter; this requirement does not apply to speed-based reselection to intra-frequency cells.
Serving Freq Low Priority RSRQ Resel Thld	NRCellReselConfig.<i>ServFreqLowPriRsrqReselThld</i>	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
High Priority Reselection RSRQ Threshold	NRCellFreqRelation.HighPriReselRsrqThld	Use the recommended value.
Low Priority Reselection Threshold	NRCellFreqRelation.LowPriReselThld	Use the recommended value.
Low Priority Reselection RSRQ Threshold	NRCellFreqRelation.LowPriReselRsrqThld	Use the recommended value.
Mobility State Evaluation Period	NRCellReselConfig.MobStateEvalPeriod	Use the recommended value.
Resel Count Threshold for Medium Mobility	NRCellReselConfig.ReselThldForMediumMob	Use the recommended value.
Resel Count Threshold for High Mobility	NRCellReselConfig.ReselThldForHighMob	Use the recommended value.
Resel Time Scale Factor for Medium Mobility	NRCellReselConfig.ReselTimeSfForMediumMob	Use the recommended value.
Additional Hysteresis for Medium Mobility	NRCellReselConfig.AddlHystForMediumMob	Use the recommended value.
Resel Time Scale Factor for High Mobility	NRCellReselConfig.ReselTimeSfForHighMob	Use the recommended value.
Additional Hysteresis for High Mobility	NRCellReselConfig.AddlHystForHighMob	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Frequency Control Switch	NRCellFreqRelation.FrequencyCtrlSwitch	The MOB_NEIGHBOR_FREQ_FILTER_SW option of this parameter controls whether to enable non-serving frequency filtering. Use the recommended value.
gNodeB Diff Layer Parameter Group ID	gNBDiffUserExpGroup.gNBDiffLayerParameterGroupId	Set this parameter based on the network plan. If this parameter and the gNodeB Frequency Priority Group ID parameter are set to the same value, the two groups respectively identified by these two parameters are associated with each other.
gNodeB Diff User Experience Group ID	NRCellOpPolicy.gNBDiffUserExpGroupId	Set this parameter based on the network plan.
gNodeB Frequency Priority Group ID	gNBFreqPriorityGroup.gNBFreqPriorityGroupId	Set this parameter based on the network plan.
Cell Reselection Priority	gNBFreqPriorityGroup.CellReselPriority	Set this parameter based on the network plan.
Cell Reselection Sub-Priority	gNBFreqPriorityGroup.CellReselSubPriority	Set this parameter based on the network plan.
T320 Timer For Other	NRCellReselConfig.T320ForOther	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Idle Reselection Policy Switch	NRCellMobilityConfig.IdleReselPolicySw	The DIFF_UE_RESEL_PRI_SUPPR_SW option of this parameter controls cell reselection priority differentiation suppression. Select this option to disable UE-layering-based differentiated cell reselection priority configuration.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

TDD configuration examples

```
//Optimizing cell reselection parameters
MOD NRCELLRESELCONFIG: NrCellId=0, MinimumRxLevel=-64, IntraFreqMeasStartThld=29,
CellReselHysteresis=DB4, CellReselPriority=7, CellReselSubPriority=0DOT0, SsbConsolidationThld=70,
SsbNumToAverage=8, NonIntraFreqMeasRsrqThld=9, NonIntraFreqMeasRsrqThld=2,
ServFreqLowPriRsrqReselThld=7, SibOptionalleInd=RSRQ_BASED_RESEL_SW-1,
ServFreqLowPriRsrqReselThld=1;
//Optimizing cell selection parameters
MOD NRDUCELLSELCONFIG: NrDuCellId=0, MinimumRxLevel=-64, SibOptionalleInd=SIB1_RSRQ_SW-1,
MinimumRxQty=-17;
//Optimizing cell relationship parameters
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, NCellReselOffset=DB0;
//Modifying the frequency relationship configurations of an NR cell
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=6300, FrequencyBand=N41, CellReselPriority=1,
CellReselSubPriority=0DOT0, MinimumRxLevel=-64, HighPriReselThld=11, LowPriReselThld=11,
FreqOffset=DB0, MinimumRxQty=-17, HighPriReselRsrqThld=3, LowPriReselRsrqThld=3;
//Modifying the SSB combination threshold and the number of SSBs allowed for calculating the average
value for inter-frequency neighboring cell measurement
MOD NRCELLMOBILITYCONFIG: NrCellId=0, SsbConsolidationThld=70, SsbNumToAverage=8;
```

```
//Turning on the protocol IE optimization switch
MOD NRCELLALGOSWITCH: NrCellId=0, HoCompatibilitySwitch=PROTOCOL_IE_OPT_SW-1;
//Adding cell blacklists for the gNodeB
ADD GNBCELLBLACKLIST: CellBlacklistId=1, StartPhysicalCellId=231, PhysicalCellIdRange=N4, RatType=NR;
ADD GNBCELLBLACKLIST: CellBlacklistId=2, StartPhysicalCellId=1, PhysicalCellIdRange=N8, RatType=NR;
//Associating the intra-frequency excluded cell list ID
MOD NRCELLMOBILITYCONFIG: NrCellId=0, IntraFExcludedCellListId=1;
//Associating the inter-frequency excluded cell list ID
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=6300, FrequencyBand=N41,
InterFExcludedCellListId=2;
//Specifying the threshold for the signal quality rankings of candidate cells in cell reselection
MOD NRCELLRESELCONFIG: NrCellId=0, CandidateCellThld=DB3;
//Configuration for enabling speed-based cell reselection
//Setting the UE mobility state evaluation period, threshold expressed as a number of cell reselections for
evaluating whether UEs enter the medium-mobility state, threshold expressed as a number of cell
reselections for evaluating whether UEs enter the high-mobility state, scaling factor for the cell reselection
time of medium-mobility UEs, scaling factor for the cell reselection time of high-mobility UEs, additional
hysteresis for cell reselection by medium-mobility UEs, and additional hysteresis for cell reselection by high-
mobility UEs
MOD NRCELLRESELCONFIG: NrCellId=0, MobStateEvalPeriod=S60, ReselThldForMediumMob=4,
ReselThldForHighMob=8, ReselTimeSfForMediumMob=1DOT0, ReselTimeSfForHighMob=0DOT75,
AddlHystForMediumMob=DB0, AddlHystForHighMob=DB0;
//Enabling speed-based cell reselection
MOD NRCELLRESELCONFIG: NrCellId=0, SibOptionalleInd=SPEED_BASED_RESEL_SW-1;
//Enabling speed-based cell reselection to inter-frequency or inter-RAT neighboring cells
MOD NRCELLRESELCONFIG: NrCellId=0, SibOptionalleInd=INTER_FREQ_RAT_SPEED_RESEL_SW-1;
//Enabling non-serving frequency filtering
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=6300, FrequencyBand=N41,
FrequencyCtrlSwitch=MOB_NEIGHBOR_FREQ_FILTER_SW-1;
//Configuring the T320 timer
MOD NRCELLRESELCONFIG: NrCellId=0, T320ForOther=MIN120;
//Enabling QCI-specific differentiated cell reselection priority configuration
//Disabling cell reselection priority differentiation suppression
MOD NRCELLMOBILITYCONFIG: NrCellId=0, IdleReselPolicySw=DIFF_UE_RESEL_PRI_SUPPR_SW-0;
//Configuring the dedicated cell reselection priority of an NR frequency for a UE
ADD GNBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=0, RatType=NR, CellReselPri=7,
CellReselSubPri=0DOT0, SsbFreqPos=6300;
//Associating the dedicated cell reselection priority with the UE's QCI
ADD GNBDIFFUSEREXPGROUP: gNBDiffUserExpGroupId=0, PolicyIndex=0, UserGradeByQci=70,
gNBCloudXParamGroupId=255, gNBDiffLayerParaGroupId=0;
//Configuring parameters related to the operator-specific policy of the NR cell
ADD NRCELLPOLICY: NrCellId=0, OperatorId=0, gNBDiffUserExpGroupId=0;
```

Deactivation Command Examples

MML commands are not required for deactivating this function. You can determine whether to restore the settings of other parameters based on actual network conditions.

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Basic Functions of Mobility Management in Idle Mode

After the related configurations are complete on the gNodeB, check SIB2 or SIB4 in the NR cell.

- Step 1** Start Uu signaling tracing: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
- Step 2** Check whether the *cellReselectionPriority* and *CellReselectionSubPriority* IEs corresponding to the NG-RAN cell are contained in SIB2 sent by the gNodeB. If they are contained and share the same values as the **NRCellReselConfig.CellReselPriority** and **NRCellReselConfig.CellReselSubPriority** parameters, respectively, cell reselection has taken effect.
- Step 3** In inter-frequency networking, check whether the *cellReselectionPriority* and *CellReselectionSubPriority* IEs corresponding to a non-serving frequency are contained in SIB4 sent by the gNodeB. If they are contained and share the same values as the **NRCellFreqRelation.CellReselPriority** and **NRCellFreqRelation.CellReselSubPriority** parameters, respectively, inter-frequency cell reselection has taken effect.

----End

 NOTE

SIB2 and SIB4 can be observed in the signaling tracing result over the Uu interface only when the content in SIB2 and SIB4 changes or the cell is activated.

Optimization to Mobility Management in Idle Mode

Blacklisted cell configuration

- Intra-frequency blacklisted cell configuration
After the related configurations are complete on the gNodeB, check SIB3 in the current serving cell by performing the following steps:
 - a. Start Uu signaling tracing: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
 - b. If the gNodeB-delivered SIB3 contains the *intraFreqExcludedCellList* IE and the cells indicated in the IE are on the cell blacklist associated with the **NRCellMobilityConfig.IntraFExcludedCellListId** parameter, the intra-frequency blacklisted cell configuration function has taken effect.
- Inter-frequency blacklisted cell configuration
After the related configurations are complete on the gNodeB, check SIB4 in the current serving cell by performing the following steps:
 - a. Start Uu signaling tracing: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
 - b. If the gNodeB-delivered SIB4 contains the *InterFreqExcludedCellList* IE and the cells indicated in the IE are on the cell blacklist associated with the **NRCellFreqRelation.InterFExcludedCellListId** parameter, the inter-frequency blacklisted cell configuration function has taken effect.

Optimized configuration for deriveSSB-IndexFromCell

After the related configurations are complete on the gNodeB, check SIB2 or SIB4 in the current serving cell.

1. Start Uu signaling tracing: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
2. If the serving cell is an NR TDD cell and the value of the *deriveSSB-IndexFromCell* IE in SIB2 sent by the gNodeB is **TRUE**, optimized configuration for deriveSSB-IndexFromCell has taken effect.
3. In inter-frequency networking, if the value of the *deriveSSB-IndexFromCell* IE corresponding to an NR TDD frequency in SIB4 sent by the gNodeB is **TRUE**, optimized configuration for deriveSSB-IndexFromCell has taken effect.

Speed-based cell reselection

After the related configurations are complete on the gNodeB, check SIB2, SIB4, or SIB5 in the current serving cell.

1. Start Uu signaling tracing: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
2. When the speed-based cell reselection function is enabled by selecting the **SPEED_BASED_RESEL_SW** option of the **NRCCellReselConfig.SibOptionalleInd** parameter, this function has taken effect if the gNodeB-delivered SIB2 contains the IEs *t-Evaluation*, *n-CellChangeMedium*, *n-CellChangeHigh*, *sf-Medium* in *q-HystSF*, *sf-High* in *q-HystSF*, *sf-Medium* in *SpeedStateScaleFactors*, and *sf-High* in *SpeedStateScaleFactors*, with their values the same as the settings of parameters **NRCCellReselConfig.MobStateEvalPeriod**, **NRCCellReselConfig.ReselThldForMediumMob**, **NRCCellReselConfig.ReselThldForHighMob**, **NRCCellReselConfig.AddlHystForMediumMob**, **NRCCellReselConfig.AddlHystForHighMob**, **NRCCellReselConfig.ReselTimeSfForMediumMob**, and **NRCCellReselConfig.ReselTimeSfForHighMob**, respectively.
3. In inter-frequency or inter-RAT networking, when the **INTER_FREQ_RAT_SPEED_RESEL_SW** option of the **NRCCellReselConfig.SibOptionalleInd** parameter is selected, the functioning of speed-based cell reselection to inter-frequency or inter-RAT neighboring cells has taken effect if the gNodeB-delivered SIB4 or SIB5 contains the IEs *sf-Medium* and *sf-High* in *SpeedStateScaleFactors*, with their values the same as the settings of parameters **NRCCellReselConfig.ReselTimeSfForMediumMob** and **NRCCellReselConfig.ReselTimeSfForHighMob**, respectively.

Activation verification is not required for other subfunctions.

4.4.3 Network Monitoring

This function is a basic function that ensures basic service performance. Separate monitoring is not required.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.304: "NR; User Equipment (UE) procedures in Idle mode and RRC"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *5G Networking and Signaling*
- *EPC-based NSA Basics*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*
- *NSA-SA Selection Based on User Experience*
- *Flexible User Steering*
- *Multi-Operator Sharing*
- *NSA Mobility Management*
- *SA Mobility Management in Connected Mode*
- *SA Mobility Management in Inactive Mode*
- *Idle Mode Management in eRAN Feature Documentation*
- *Mobility Management in Connected Mode in eRAN Feature Documentation*
- *FWA*
- *CA Experience Improvement*
- *Mobility Load Balancing in Idle Mode*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*